

Solution Requirements

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Solution Requirements

This document outlines both the Functional and Non-Functional Requirements defining exactly what the Emotion Detection & Learning Support Engine solution will perform and the operational standards it must meet.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system must authenticate student and administrative sessions securely using local credentials matched against database records.</i>	High
2	Authorization levels	<i>System access must be divided into discrete role clearances: standard Students who can submit text metrics and Admins/Counselors authorized to view backend dashboard charts.</i>	Medium
3	External interfaces	<i>The application must interface directly with the Google Gemini API (gemini_service.py) using secure API handlers to generate support tips.</i>	High

		<i>services the system must connect to]</i>	
4	Transactions processing	<i>The application backend must process incoming text inputs dynamically through PyTorch deep learning models (BERT/BiLSTM) to predict the primary emotion and confidence score.</i>	High
5	Reporting	<i>The platform must generate real-time visual summaries, charting emotional distributions, confidence scores, and history logs via an interactive visualization engine (dashboard.py).</i>	Medium
6	Business rules, etc.	<i>Inputs registering high anxiety values must automatically display direct support shortcuts, while general technical topics trigger tailored study guides.</i>	High
7	Compliance to laws or regulations	<i>The application must protect sensitive student information by securely hashing local user profiles and anonymizing text history logs within local storage.</i>	High
8	Other	<i>[Add any additional functional requirements here]</i>	

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	<i>The system must maintain fast inference execution loops for the deep learning modules and external API routes.</i>	<i>End-to-end text tokenization, neural classification, and support generation must complete in < 1.5 seconds.</i>
2	Scalability	<i>The application must handle sudden surges in traffic during exam periods without server performance degradation.</i>	<i>The platform must support up to 500 concurrent text classification workflows simultaneously.</i>
3	Security & Data Privacy	<i>The local data stores must protect student records from unauthorized access or exposure.</i>	<i>Hashed user authorization storage and isolated local configurations for private API credentials</i>
4	Reliability & Availability	<i>The application must keep services online consistently and provide failover paths if connections drop.</i>	<i>Maintain a minimum of 99.5% service uptime with automated context restoration parameters.</i>
5	Usability & Accessibility	<i>The front-end view layouts must be clean, simple, and responsive across varying screen layouts.</i>	<i>Complete support form rendering on both web and mobile screens within standard WCAG accessibility design guidelines.</i>